

MAC 202 Course Outline

MAC 202 — FINEX Macro-Fiscal Modelling

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	36 (indicative band 30–40)
Shape	10 modules · 15 sessions · 10 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101; MAC 201 or equivalent intermediate macroeconomics
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-22 · 12-month cycle

What this course is

MAC 202 trains analysts to construct, run, validate and interpret the EDIS FINEX macro-fiscal framework. FINEX is the AICTEM Julia port of the IMF Forecasting Model of Internal and External Balance (Berg, Hul, Karam, Remo and Rodriguez, IMF Working Paper WP/23/235): a semi-structural, annual, open-economy gap model with rational expectations, 349 equations, 83 structural shocks and three monetary-policy regimes.

The rule the course is built on. The course is built on the implemented model, not on a textbook description of one. Every equation taught is quoted from `FINEX_canonical.model`; every numerical result is committed output from the AICTEM Utopia application; every limitation is named where a learner will meet it.

Who it is for

Ministry of Finance, planning, budget, debt and sector-finance analysts

What this course does not claim

The laboratories run the FINEX engine through its own Julia and Python command line. None runs through the EDIS platform: no engine has yet been executed end to end from AICTEMIntegrated (limitation G-01), and the CLI is the reproducible, tested surface. Laboratory 08 constructs the FINEX–MTEF handover by hand from the adapter’s source, which is both honest and more instructive.

Structure

Module	Title	Sessions	Laboratory	Hours
M01	Architecture and scope	01	01	1.5
M02	Real sector and national accounts	02–03	01–02	4.8
M03	Fiscal block	04–06	03–04	7.3
M04	External and monetary	08–09	—	2.8
M05	Debt	07	05	4.4
M06	Method: inversion, SMM, validation	10–11	02, 07	3.7
M07	Scenario design	12	06	4.1
M08	Integration	13–14	08	3.6
M09	Reporting and judgement	15	07	2.0
CAP	Capstone	—	—	1.8
	Total	15	10	36.0

The 15 sessions and 10 laboratories, in teaching order, and the subject matter of each:

M01 Architecture and scope — Session 01 · 1.5 h · Laboratory 01

- **01 FINEX architecture** — Scope, the five properties, eleven blocks, naming, regimes, nonlinearity
- *Lab 01 Load and validate national macroeconomic data* — build a FINEX observable panel from institutional sources, apply the model’s own transforms, and produce a data-quality note

M02 Real sector and national accounts — Sessions 02–03 · 4.8 h · Laboratory 01–02

- **02 National accounts** — The nominal GDP identity, six components, trends, market clearing
- **03 The real sector** — Potential output, the public-capital channel, time to build, the gap
- *Lab 02 Construct a macro baseline* — solve the country steady state, estimate the behavioural coefficients and shock variances by SMM, and produce the reference baseline with its fit table

M03 Fiscal block — Sessions 04–06 · 7.3 h · Laboratory 03–04

- **04 The fiscal sector** — Four identities, determination types, the residual, sign conventions
- **05 Revenue forecasting** — Effective rates, $\frac{1}{1+}$, calibration, cyclical adjustment
- *Lab 03 Revenue forecasting and a tax reform* — decompose revenue by base, run a tax reform as a permanent instrument change, and quantify the behavioural offset the static estimate misses
- **06 Expenditure** — The wage-bill boundary, transfers, capital, interest
- *Lab 04 Expenditure and the fiscal envelope* — build the primary expenditure envelope from model output, net interest exactly once, and reconcile it against a real approved budget

M04 External and monetary — Sessions 08–09 · 2.8 h · Laboratory —

- **08 The external sector** — Four flows, the BoP identity, the premium, the crisis loop
- **09 Inflation and exchange rates** — The Phillips curve, the Taylor rule, regimes, five FX channels

M05 Debt — Session 07 · 4.4 h · Laboratory 05

- **07 Debt** — Two currencies, decomposition, the anchor, the exponential premium
- *Lab 05 Debt dynamics and the debt premium* — decompose a debt path into its four contributions, compute the debt-stabilising primary balance, and find where the premium loop stops converging

M06 Method: inversion, SMM, validation — Sessions 10–11 · 3.7 h · Laboratory 02, 07

- **10 Baseline forecasting** — The inversion, SMM, diagnostics, the three disclosures
- **11 Fiscal space** — The calculation, netting, consistency, the anchor trap
- *Lab 07 Fiscal space and the policy note* — compute fiscal space from your own run, test it, and write a one-page note that meets the eight-element result contract

M07 Scenario design — Session 12 · 4.1 h · Laboratory 06

- **12 Scenario analysis** — Shock or path, signs, anticipation, the eight families
- *Lab 06 Shock scenarios and impulse responses* — read the impulse-response table, design three scenarios using the shock/path test, run them cleanly, and report differences

M08 Integration — Sessions 13–14 · 3.6 h · Laboratory 08

- **13 FINEX and MTEF** — The variable map, netting once, MTEF’s rules, convergence
- *Lab 08 The FINEX–MTEF handover* — build the canonical envelope by hand, produce the reconciliation diagnostic, and apply a backward-feedback conditioning path
- **14 FINEX and CDCGE** — Two models, the partition, unpublished channels

M09 Reporting and judgement — Session 15 · 2.0 h · Laboratory 07

- **15 Policy reporting** — The result contract, the note, charts, forbidden sentences

CAP Capstone — Session — · 1.8 h · Laboratory —

Further laboratories:

- *Lab 09 Impulse responses: run, check, tabulate* — run the impulse-response suite for one country, verify every sign against its convention, and produce the peak table with its sources and four summary statistics
- *Lab 10 Shock decomposition and the scenario suite* — decompose four episodes with the five-step protocol, run the six-scenario suite off a common baseline, apply the eight pre-flight checks, and report to contract

What a graduate can do

1. Explain what kind of model FINEX is, what it determines, and what it deliberately does not.
2. Read any FINEX equation and identify its variables, units, coefficients, shock and steady-state behaviour.
3. Assemble and validate a country observable panel from institutional sources, applying the model’s own transforms.
4. Calibrate a country steady state and defend each anchor.
5. Estimate the behavioural coefficients and shock variances by SMM, and diagnose identification.
6. Produce a medium-term baseline using the historical inversion, and state correctly what it is.
7. Design, run and report a scenario, choosing correctly between a shock and a conditioning path.
8. Compute fiscal space, the debt-stabilising primary balance and the debt-dynamics decomposition from model output.
9. Validate a run against the eight-level hierarchy and report the diagnostics honestly.
10. Hand a FINEX envelope to MTEF in canonical form with interest netted exactly once.

The 12 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	0 items across 9 modules
Laboratory exercises	30 %	10 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	country panel and data-quality report; calibration defence; fiscal reform appraisal; debt sustainability note
Final assessment	30 %	Three hours, open book: definitions, calculation, interpretation and reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A laboratory manual, instructor guide and learner workbook; and the lms package. The README lists every file.

Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

MAC 202 FINEX Macro-Fiscal Modelling · version 1.0 · AICTEM EDIS Academy, School 02 Macroeconomics and Public Finance.